

CSCE 658 RANDOMIZED ALGORITHMS · MODULE 06 · LECTURE NOTES

Random Graphs

Phase transitions, and why they matter.

MODULE OUTCOMES

- Define the Erdős–Rényi models.
- Explain the giant component transition.
- Explain the connectivity and other thresholds.
- Explain the first and second moment methods.
- Relate random graph thresholds to real phase transitions.

1 The model and its thresholds

$G(N, p)$: N VERTICES, EACH EDGE PRESENT INDEPENDENTLY WITH PROBABILITY p

The simplest possible random graph model — and the striking fact about it is that **almost every property of interest appears suddenly at a threshold value of p rather than increasing gradually.**

At $p = 1/n$ the giant component appears. Below that, all connected components have size $O(\log n)$; above it, a single component contains a constant fraction of all n vertices. **Average degree exactly 1 is the tipping point**, which is the most famous result in the subject.

At $p = \ln n / n$ the graph becomes connected, and the last isolated vertex disappears at precisely the same moment — which is not a coincidence: the isolated vertex is the binding obstacle, and once none remains the graph is connected.

And the transitions are sharp: the window of p in which the property goes from unlikely to likely shrinks as n grows, so for large n the behaviour is effectively a step function rather than a smooth curve. That sharpness is what makes 'threshold' the right word.

Property	Threshold	Note
A giant component exists	$p = 1/n$	Average degree 1. The famous one, and the basis of percolation theory.
The graph is connected	$p = \ln n / n$	Exactly the same threshold as 'no isolated vertices' , and for the reason in the callout.
A triangle appears	$p = 1/n$	The expected number of triangles is constant at this scale , which is where the first-moment calculation turns over.

Property	Threshold	Note
A Hamiltonian cycle exists	$p = \ln n / n$	The same threshold as mere connectivity, which is surprising — see the note.
A perfect matching exists	$p = \ln n / n$	Also the same, and for a related reason.
Diameter at most 2	p about $\sqrt{(2 \ln n / n)}$	Much denser — every pair needs a common neighbour, which is a far stronger requirement.

That Hamiltonicity appears at the same threshold as connectivity is genuinely striking. The obvious obstacle to a Hamiltonian cycle is a vertex of degree less than two, and once those have all disappeared — which happens at the connectivity threshold — the cycle exists with high probability. So the only obstruction is the local one, which is not at all what you would expect given that Hamiltonicity is NP-complete in general (CSCE 637 Module 04 §2).

2 The first and second moment methods

TO SHOW A STRUCTURE DOES NOT APPEAR (below threshold):

FIRST MOMENT METHOD.

Let X = the number of copies of the structure.
 If $E[X] \rightarrow 0$, then $P(X \geq 1) \leq E[X] \rightarrow 0$,
 by Markov's inequality (Module 02 section 1).
 So with high probability there are NONE.

TO SHOW IT DOES APPEAR (above threshold):

SECOND MOMENT METHOD.

$E[X] \rightarrow \infty$ is NOT sufficient -- the expectation
 could be carried by rare instances with enormous X ,
 while most instances have $X = 0$.
 But if $\text{Var}[X] / E[X]^2 \rightarrow 0$, then Chebyshev gives
 $P(X = 0) \rightarrow 0$.
 So with high probability there is at least one.

THAT IS THE WHOLE TOOLKIT for threshold results, and it is
 nothing more than Markov and Chebyshev (Module 02).

THE ASYMMETRY IS THE POINT: showing a structure is ABSENT
 needs only the mean; showing it is PRESENT needs the
 variance as well. Which is why the second moment
 calculation is the hard half of every such proof, and why
 lower-bound and upper-bound halves of a threshold result
 have completely different difficulty.

Absence needs the mean; presence needs the variance — and that asymmetry recurs throughout probabilistic combinatorics and into Module 07's probabilistic method, where the same two-sided structure appears. It is the transferable insight from this module, considerably more so than any particular threshold value.

3 Real phase transitions

- **Random SAT.** Random 3-SAT is easy below a clause-to-variable ratio of about 4.27 (because solutions are abundant) and easy above it (because unsatisfiability is easy to establish), and hard right at it — which is CSCE 637 Module 12's parameter, and it is a genuine sharp transition with the hard instances concentrated in a narrowing window.
- **Percolation.** Connectivity in a random network, which is the standard model for epidemic spread, forest fire propagation, and material fracture — and the giant component threshold is the epidemic threshold.
- **Network robustness.** Remove a random fraction of nodes and the giant component vanishes at a threshold — which is how network resilience is analysed (CSCE 678), and the threshold differs sharply between random failure and targeted attack on high-degree nodes.
- **Cuckoo hashing's load factor** (Module 04 §3) is a threshold in a random bipartite graph — the insertion cascade terminates exactly when that graph has no dense subgraph, which happens below 0.5.
- **And community detection's detectability threshold** — below a certain signal strength in a stochastic block model, no algorithm whatsoever can recover the communities, provably. Which is the most striking item here: an information-theoretic limit below which no amount of computation helps, and that is a different kind of impossibility from CSCE 627's — the information is absent rather than the computation infeasible.

WHY REAL NETWORKS ARE NOT ERDŐS-RÉNYI

Real networks have heavy-tailed degree distributions, high clustering, and community structure. $G(n,p)$ has none of those — its degree distribution is Poisson (thin-tailed), its clustering coefficient tends to zero, and it has no communities by construction.

So the thresholds are not quantitatively predictive for a real network, and the models that are — preferential attachment, the configuration model, stochastic block models — have different thresholds, sometimes dramatically so (scale-free networks have no epidemic threshold at all in the infinite limit).

But the existence of sharp transitions transfers — real networks also exhibit thresholds, and §2's moment methods are how they are located in whichever model is appropriate.

So use $G(n,p)$ as the null model. 'A random graph with this degree sequence would have a clustering coefficient of 0.001 and we observe 0.3' is the useful form — the same baseline-not-prediction use as Module 03 §4, and the pattern is worth noticing: the simple random model's value is as a comparison point, not as a description.

4 Random graphs as a construction tool

- **Expander graphs.** A random regular graph is an expander with high probability, and expanders are the central object in derandomisation (Module 12 §3) — so the existence proof is probabilistic and the explicit constructions came decades later and are harder.

- **Error-correcting codes.** Random sparse parity-check matrices give codes approaching the channel capacity (LDPC codes), which is a probabilistic existence proof with a very large practical consequence — these codes are in every modern wireless standard.
- **Testing and benchmarking.** Random instances are the standard benchmark for graph algorithms — with the important caveat that they are frequently much easier than real instances (CSCE 637 Module 01 §3), because real instances have structure that random ones lack.
- **Network design.** Random regular graphs have small diameter and good fault tolerance for their degree, which is why some datacentre network topologies are deliberately randomised (Jellyfish and its successors) rather than structured.
- **And the probabilistic method,** which is Module 07 and is the general form of all four of these. 'A random object has the property, therefore an object with the property exists' is the move, and random graphs are its most productive setting.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 5 and 6	Random graphs and the moment methods of §2, at the right level for this course.
Bollobás — Random Graphs	The comprehensive reference for §1's thresholds. Library copy.
Newman — Networks	§3's real networks, including why they differ from $G(n,p)$ and which models fit. Library copy; the author's lecture notes are free.
Decelle, Krzakala, Moore & Zdeborová — the detectability threshold (free)	§3's most striking result — the information-theoretic limit on community detection.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Generate $G(n,p)$ for a range of p and plot the largest component size.** Locate the transition.
- 2 **Confirm the transition sharpens as n grows.**
- 3 **Plot the connectivity probability** against p and locate $\ln n / n$.
- 4 **Count isolated vertices** and confirm they vanish at the same point.
- 5 **Test for Hamiltonicity** at several p and confirm it coincides with connectivity.
- 6 **Apply the first moment method** to show triangles are absent below the threshold.

- 7 **Apply the second moment method** to show they are present above it, and note which half was harder.
- 8 **Generate random 3-SAT at several clause ratios** and plot solver time. Locate 4.27.
- 9 **Compute the clustering coefficient** of a real network and of $G(n,p)$ with matching density.
- 10 **Generate a random regular graph** and measure its diameter against a structured topology of the same degree.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define $G(n,p)$ and state the striking fact about its properties.
- 2 Give six thresholds and what appears at each.
- 3 Why do connectivity and 'no isolated vertices' coincide?
- 4 Why is the Hamiltonicity threshold surprising?
- 5 State the first and second moment methods and the inequality each uses.
- 6 Why is the asymmetry between them the point?
- 7 Name five real phase transitions.
- 8 Which is information-theoretic rather than computational, and why does that differ?
- 9 Why are real networks not $G(n,p)$, and how should the model be used?
- 10 Name four constructive uses of random graphs.